

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Thomas Kwa, Ben West, Joel Becker, et al. — METR (Model Evaluation & Threat Research)

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~7 months

50% time-horizon
doubling period

$R^2 = 0.97$

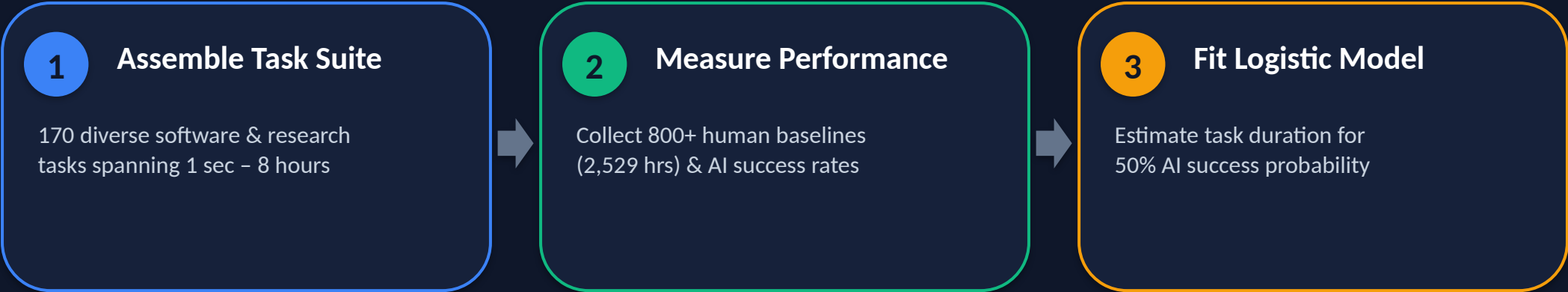
Exponential fit over
6 years (2019–2025)

2 sec → ~2 hrs

GPT-2 (2019) to
o3 (2025)

A New Metric: 50% Task-Completion Time Horizon

The task duration (measured by skilled human completion time) at which an AI model achieves 50% autonomous success rate — an intuitive, human-grounded measure of real-world AI capability.



Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST • 97 tasks

Diverse software tasks

Range: 1 min – 30 hrs

RE-Bench • 7 tasks

ML research engineering

Range: 8 hrs each

SWAA • 66 tasks

Single-step atomic actions

Range: 1 sec – 30 sec

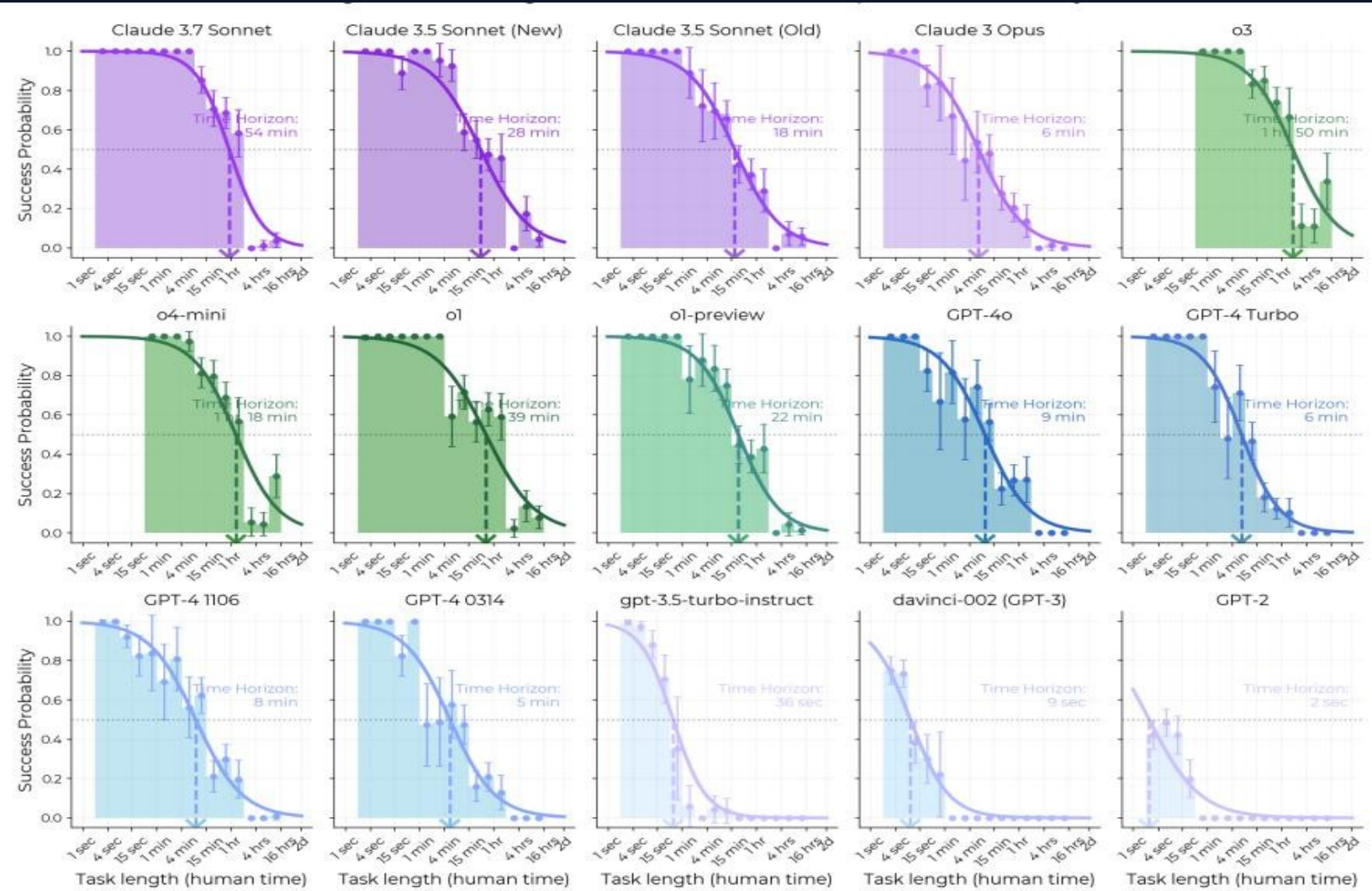
800+ human baselines

2,529 hours collected

Task Family	Human Time	Description	Source
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"	SWAA
wikipedia_research	1 minute	Research simple factual information from Wikipedia	HCAST
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files	HCAST
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules	HCAST
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool	RE-Bench

AI Task Horizon: 2 Seconds (GPT-2) → ~2 Hours (o3) in 6 Years

Doubling time ~207 days (~7 months) Fit quality $R^2 = 0.97$ 2023–2025 rate ~20% faster than 2019–2025

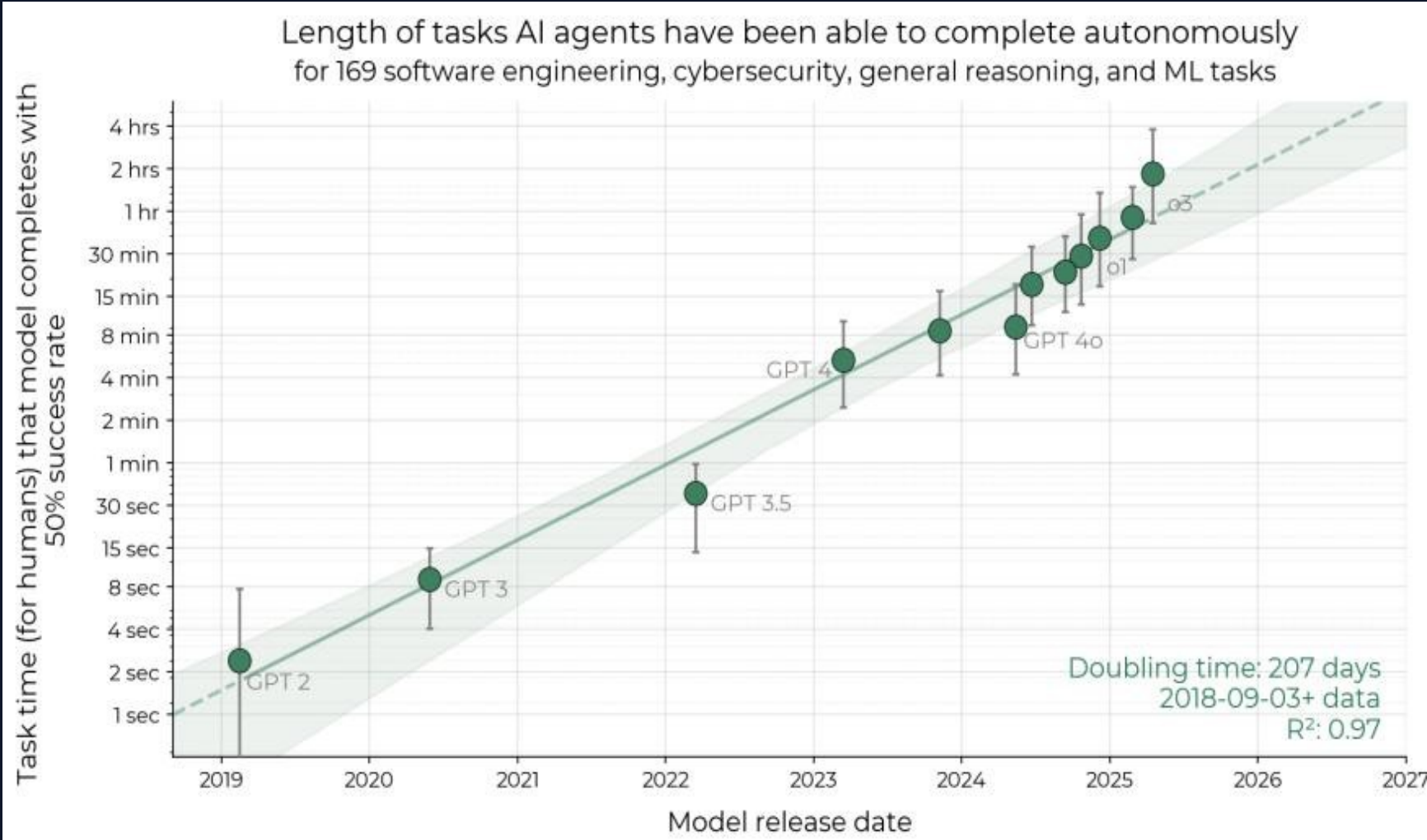


Key Model Milestones

GPT-2	2019	~2 sec
GPT-3	2020	~9 sec
GPT-3.5	2023	~36 sec
GPT-4 (0314)	2023	~5 min
GPT-4o	2024	~9 min
Claude 3 Opus	2024	~6 min
Claude 3.5 Sonnet	2024	~28 min
o1	2024	~39 min
Claude 3.7 Sonnet	2025	~54 min
o3	2025	~1 hr 50 min

Success Probability Drops Sharply as Task Length Increases

Each curve shows a model's success rate vs. human-measured task duration. Frontier models push the curve rightward — the 80% time horizon tracks ~5× shorter than the 50% horizon.



o3

~1 hr 50 min

Current frontier for 50% success

Claude 3.7

~54 min

Strong on structured software tasks

GPT-2

~2 sec

Baseline — near-zero beyond trivial tasks

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four capability dimensions underpin the exponential growth in autonomous task completion:

Improved Logical Reasoning

Models exhibit stronger multi-step deduction, hypothesis testing, and planning over longer task horizons.

Better Tool Use

Frontier models interact more effectively with code editors, terminals, APIs, and file systems — chaining tool calls across complex workflows.

Greater Reliability & Self-Awareness

Modern agents recognize failure states, request clarifications, and avoid cascading errors through improved introspection.

Ability to Adapt to Mistakes

Error recovery has improved dramatically — agents can diagnose bugs, revert faulty changes, and try alternative strategies autonomously.

Key Limitation

Performance is notably lower on less structured, "messier" tasks. External validity to real-world job distributions remains an open question.

Extrapolation: Month-Long Task Automation by 2028–2031

Naïve extrapolation of the exponential trend projects when AI agents could autonomously complete tasks requiring 167 work hours (~1 month of human effort).

Projection Scenarios

Mid-2028

Accelerated trend (2023–2025 rate)

20% faster growth continues as reasoning capabilities compound

~2029–2030

Baseline trend (2019–2025 rate)

Doubling every ~207 days continues at current exponential pace

Mid-2031

Decelerated trend

Growth slows due to task complexity ceiling or data limitations

Critical Thresholds



~2 hours

Current frontier (o3, 2025)

Today



~1 day (8 hrs)

Full workday automation

~2026



~1 week (40 hrs)

Sprint-length tasks

~2027



~1 month (167 hrs)

Project-length automation

2028–2031

Caveat: Extrapolation assumes continued exponential scaling. Real-world task distributions, compute constraints, and regulatory factors may alter trajectories.

Takeaways: Exponential Capability Growth Demands Proactive Safety Governance

1 Robust, Intuitive Metric

The 50% task-completion time horizon provides a human-grounded benchmark for tracking autonomous AI capability growth, achieving $R^2 = 0.97$ over 6 years of model releases.

2 Doubling Every ~7 Months, Possibly Accelerating

The ~207-day doubling time is consistent across the full 2019–2025 period, with the 2023–2025 sub-period showing ~20% faster growth — suggesting acceleration, not saturation.

3 Current Frontier: ~2 Hours of Autonomous Work

o3 can autonomously complete tasks that take skilled humans nearly 2 hours. This represents a ~3,000× increase over GPT-2's 2-second horizon in just 6 years.

4 External Validity Remains an Open Question

The 170-task suite does not perfectly represent the full distribution of real-world intellectual labor. Supplementary experiments found little evidence of slower trends on realistic tasks, but coverage gaps persist.

5 Unstructured Tasks and Generalization Are Key Limitations

Performance degrades on less structured, "messier" tasks. Whether the exponential trend holds as AI tackles more ambiguous, open-ended work is a critical unknown for safety planning.